

2018

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**



Hunters Hill
High School

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen only
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 14 show relevant mathematical reasoning and/or calculations

Total Marks – 70

Section I Questions 1 – 10 **10 marks**

Allow about 15 minutes for this section

Section II Questions 11 – 14 **60 marks**

Allow about 1 hour and 45 minutes for

Name: _____

Section 1 10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1 Which of the following expressions is equivalent to $\cos x + \sqrt{3} \sin x$?

(A) $2\cos\left(x + \frac{\pi}{3}\right)$

(B) $2\cos\left(x - \frac{\pi}{3}\right)$

(C) $2\cos\left(x + \frac{\pi}{6}\right)$

(D) $2\cos\left(x - \frac{\pi}{6}\right)$

2 If $(x - 2)$ and $(x + 1)$ are factors of $x^3 + x^2 + bx + c$ what is the value of $(b + 2c)$?

(A) -4

(B) 12

(C) -8

(D) -12

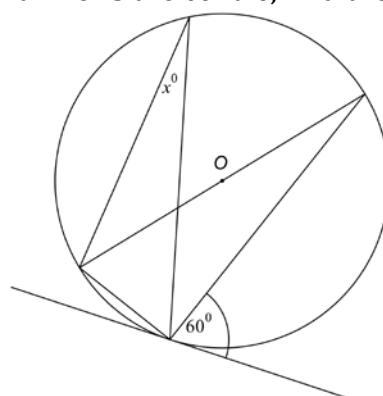
3 The diagram below shows a circle with tangent. If O is the centre, find the value of x° .
Diagram is not to scale.

(A) 60°

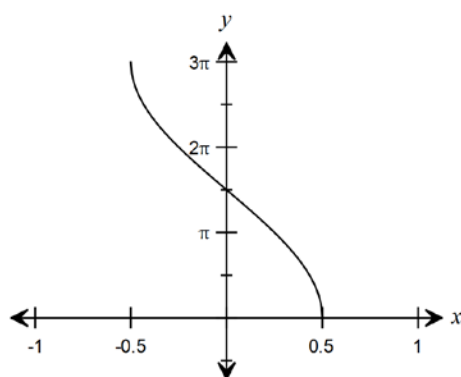
(B) 30°

(C) 15°

(D) 20°



- 4 Which of the following equations represents the graph below?



- (A) $y = 2 \cos^{-1}(3x)$
 (B) $y = 2 \left| \sin^{-1}(3x) \right|$
 (C) $y = 3 \cos^{-1}(2x)$
 (D) $y = 3 \sin^{-1}(2x)$
- 5 Evaluate $\int_0^1 \frac{e^x}{1 + e^x} dx$.

- (A) $\frac{e}{1 + e}$
 (B) $\frac{e^2}{1 + e^2}$
 (C) $\ln(1 + e)$
 (D) $\ln\left(\frac{1 + e}{2}\right)$

- 6 A parabola has the parametric equations $x = 6t$, $y = 3t^2$. Hence $\frac{dy}{dx}$, in Cartesian form, is equal to which of the following?

- (A) $6t$
 (B) $\frac{x^2}{12}$
 (C) $\frac{x}{6}$
 (D) $\frac{2x}{9}$

7 What is the general solution of $\cos 2\alpha = \frac{1}{\sqrt{2}}$

(A) $\alpha = \frac{\pi}{8} + n\pi$ or $\alpha = -\frac{\pi}{8} + n\pi$, for $n \in \mathbb{Z}$.

(B) $\alpha = \frac{\pi}{8} + 2n\pi$ or $\alpha = \frac{7\pi}{8} + 2n\pi$, for $n \in \mathbb{Z}$.

(C) $\alpha = \frac{\pi}{4} + n\pi$ or $\alpha = \frac{3\pi}{4} + n\pi$, for $n \in \mathbb{Z}$.

(D) $\alpha = \frac{\pi}{4} + 2n\pi$ or $\alpha = \frac{3\pi}{4} + 2n\pi$, for $n \in \mathbb{Z}$.

8 A bag contains 5 identical blue marbles, 6 identical black marbles and 3 identical red marbles. Three marbles are drawn at random. Which expression below gives the correct probability that exactly two blue marbles are drawn?

(A) $\frac{{}^5C_2}{{}^{14}C_3}$

(B) $\frac{{}^5C_2 \times {}^9C_1}{{}^{14}C_3}$

(C) $\frac{2}{5} \times \frac{1}{9}$

(D) $\frac{2}{14} \times \frac{1}{9}$

- 9 If $f(x) = \frac{3+e^{2x}}{5}$, which of the following is $f^{-1}(x)$?
- (A) $\ln(5x-3)$
 - (B) $\frac{1}{2}\ln(5x-3)$
 - (C) $\ln(5x)-\ln(3)$
 - (D) $\frac{1}{2}(\ln(5x)-\ln(3))$
- 10 The population, P , of animals in an environment in which there are scarce resources is increasing such that $\frac{dP}{dt} = P(100-P)$, where t is time. The initial population is 20 animals. Which of the following is true?
- (A) $P = 100 - 80e^{100t}$
 - (B) The population is increasing most rapidly when $P = 50$.
 - (C) The population is increasing most rapidly when $t = 50$.
 - (D) The maximum population is $P = 50$.

End of Section 1

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE page. Extra writing booklets are available.

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Determine the acute angle between the lines $x + y - 5 = 0$ and $x - 2y - 5 = 0$, correct to the nearest minute. 2
- (b) Evaluate $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \cos^2 2x \, dx$ leaving your answer in exact form. 3
- (c) (i) Sketch $y = \frac{x-1}{x+1}$, showing all asymptotes and intercepts. 3
- (ii) Hence, or otherwise, solve $\frac{x-1}{x+1} > 1$. 2
- (d) a, b and c are the roots of the polynomial $3x^3 + 4x^2 - 5x - 8 = 0$
- Find the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$. 2
- (e) Find the exact value of $\sin 2 \left(\tan^{-1} \frac{1}{2} \right)$ 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.
- (i) Find the coefficient of x^6 . 2
- (ii) Determine the size of the greatest coefficient. 2
- (b) Use the substitution $u = 9 - x$ to evaluate $\int_0^5 x\sqrt{9-x} \, dx$ 3
- (c) The point $P(1,2)$ divides the interval AB in the ratio $k:1$. If A is the point $(-3,6)$ and B is the point $(7,-4)$, determine the value of k ? 2
- (d) Two metals, A and B , are heated in separate ovens. Metal A is heated to a temperature of 175° and metal B is heated to a temperature of 275° . The metals are taken out of the ovens at the same time and left in a room to cool. The metals cool at different rates. The temperature of metal A is given by $T_A = 25 + 150e^{-\left(\frac{1}{20} \ln \frac{3}{2}\right)t}$ where t is the time in minutes after the metals have been removed from the ovens. The temperature of metal B is given by $T_B = 25 + 250e^{-kt}$
- (i) Twenty minutes after being removed from the oven the temperature of metal B is 175° . Show that 1
- $$k = \frac{1}{20} \ln \frac{5}{3}.$$
- (ii) How many minutes after being removed from the ovens will the metals have the same temperature? Write your answer to the nearest minute. 2
- (e) Prove by mathematical induction that
- $$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$
- n is any positive integer, where $n \geq 1$. 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Sketch $f(x) = \sin^{-1}\left(\frac{5x}{3}\right)$. 2

(ii) Find $f'(x)$ for the function. 2

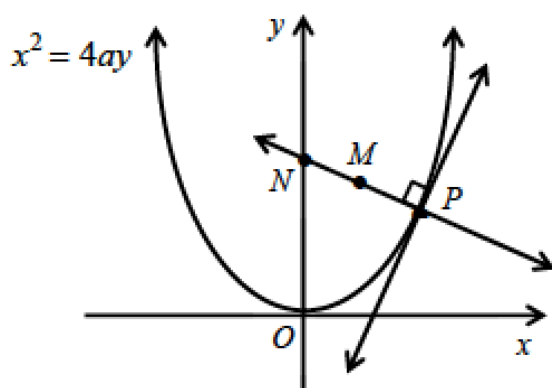
(b) Evaluate $\int_0^{\frac{1}{\sqrt{3}}} \frac{2}{9+4x^2} dx$. 3

(c) An office network consisting of 20 computers has been attacked by a computer virus. The probability that any particular computer has been affected by the virus is $\frac{1}{5}$. A computer technician has to check each computer individually. 3

(i) Write down an expression for the probability that exactly six of the computers have been affected.. 1

(ii) Find the probability that exactly six of the computers have been affected and they are the first six computers that the technician checks. 2

(d) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$. The normal to the parabola at P cuts the y -axis at N . M is the midpoint of PN .



(i) Show that N has coordinates $(0, ap^2 + 2a)$. 1

(ii) Determine the coordinates of M . 1

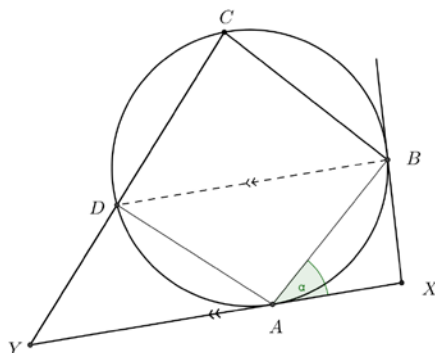
(iii) Hence, find the locus of M as P moves on the parabola. 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) If $3\sin x + \sqrt{3}\cos x \equiv R\sin(x + \alpha)$, determine the values of R and α . **2**

(b)



The diagram above shows the cyclic quadrilateral $ABCD$. The tangents drawn from point X touch the circle at A and B . XA produced meets CD produced at Y . The chord DB is parallel to YX .

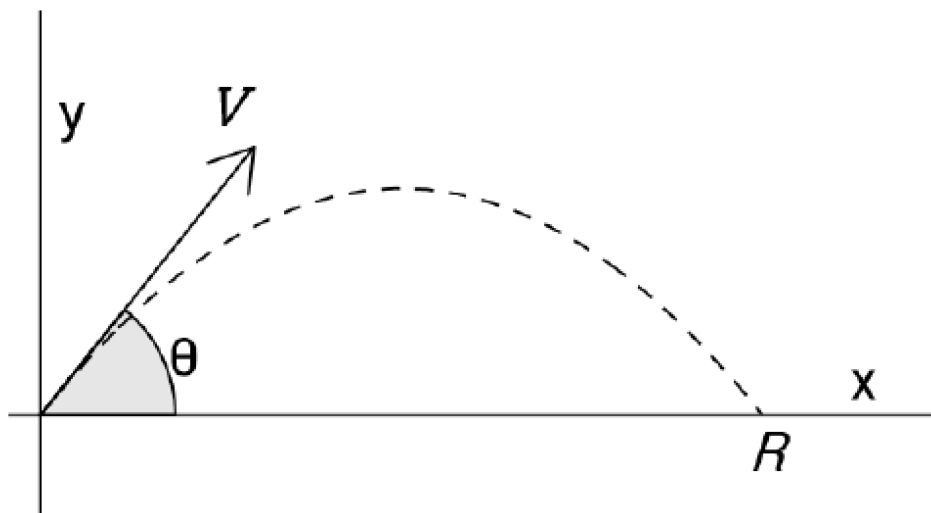
- (i) Given $\angle BAX = \alpha$ show that $\angle BCD = 2\alpha$. **3**
- (ii) Show that $BXYC$ is a cyclic quadrilateral. **2**
- (c) A four letter employee password is formed from the letters A, B, C, D, E and F .
- (i) If repetition of letters is not allowed, how many different passwords beginning with A can be formed? **1**
- (ii) If repetition is allowed, how many different passwords can be formed if exactly one of the letters must appear twice? **2**

Question 14 continues on page 16

Question 14 (continued)

- (d) The projectile is fired on the x-y plane with initial velocity V and an angle of projection θ . You are given that the Cartesian equation of the projectile is

$$y = \tan\theta x - \frac{gx^2}{2V^2} \sec^2\theta \quad (\text{Do not prove this.})$$



- (i) Show that the range is given by $R = \frac{V^2 \sin 2\theta}{g}$. **2**
- (ii) A projectile falls 100 metres short of a target when the angle of projection is 15° and lands 558.8 metres past the target when the angle of projection is 30° . Find the angle of projection required to hit the target giving your answer correct to the nearest minute. **3**

End of Paper.

DRAFT
SOLUTIONS

Name: _____

2018
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

Multiple-Choice Answer Sheet

Select the alternative A, B, C, or D that best answers the question by placing an X in the box.

	A	B	C	D
1		X		
2				X
3		X		
4			X	
5				X
6			X	
7	X			
8		X		
9		X		
10		X		

Section 1 10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 - 10.

1 Which of the following expressions is equivalent to $\cos x + \sqrt{3} \sin x$?

(A) $2 \cos \left(x + \frac{\pi}{3} \right)$

(B) $2 \cos \left(x - \frac{\pi}{3} \right)$

(C) $2 \cos \left(x + \frac{\pi}{6} \right)$

(D) $2 \cos \left(x - \frac{\pi}{6} \right)$

$R = 2$
 $\text{phase} = \sqrt{3}$
 $\alpha = \frac{\pi}{3}$

(B)

2 If $(x-2)$ and $(x+1)$ are factors of $x^3 + x^2 + bx + c$ what is the value of $(b+2c)$?

(A) -4

(B) 12

(C) -8

(D) -12

$8 + c + 2b + c = 0$
 $2b + c = -12$ --- (1)
 $-1 + 1 - b + c = 0$
 $-b + c = 0$ --- (2)
 $b + 2c = (-12)$
 (D)

3

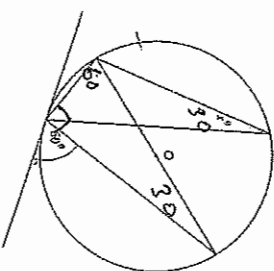
The diagram below shows a circle with tangent. If O is the centre, find the value of x° .
Diagram is not to scale.

(A) 60°

(B) 30°

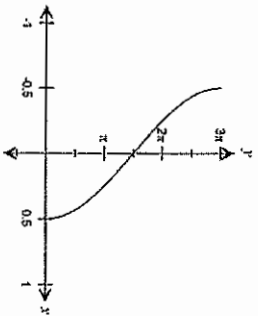
(C) 15°

(D) 20°



(B)

4 Which of the following equations represents the graph below?



- (A) $y = 2 \cos^{-1}(3x)$
 (B) $y = 2 |\sin^{-1}(3x)|$
 (C) $y = 3 \cos^{-1}(2x)$
 (D) $y = 3 \sin^{-1}(2x)$

or
 $0 \leq \frac{y}{3} \leq \pi$
 $0 \leq y \leq 3\pi$
 $-1 \leq 2x \leq 1$
 $-0.5 \leq x \leq 0.5$

C

5 Evaluate $\int_0^1 \frac{e^x}{1+e^x} dx$.

- (A) $\frac{e}{1+e}$
 (B) $\frac{e^2}{1+e^2}$
 (C) $\ln(1+e)$
 (D) $\ln\left(\frac{1+e}{2}\right)$

$\int \frac{e^x}{1+e^x} dx$
 $= \ln|1+e^x| + C$
 $= \ln|1+e| - \ln|1+1|$
 $= \ln\left(\frac{1+e}{2}\right)$

D

6 A parabola has the parametric equations $x = 6t$, $y = 3t^2$. Hence $\frac{dy}{dx}$, in Cartesian form, is equal to which of the following?

- (A) $6t$
 (B) $\frac{x^2}{12}$
 (C) $\frac{x}{6}$
 (D) $\frac{2x}{9}$

$t = \frac{x}{6}$
 $\therefore y = 3\left(\frac{x}{6}\right)^2$
 $= \frac{x^2}{12}$
 $\frac{dy}{dx} = \frac{x}{6}$

C

7 What is the general solution of $\cos 2\alpha = \frac{1}{\sqrt{2}}$?

- (A) $\alpha = \frac{\pi}{8} + n\pi$ or $\alpha = -\frac{\pi}{8} + n\pi$, for $n \in \mathbb{Z}$.
 (B) $\alpha = \frac{\pi}{8} + 2n\pi$ or $\alpha = \frac{7\pi}{8} + 2n\pi$, for $n \in \mathbb{Z}$.
 (C) $\alpha = \frac{\pi}{4} + n\pi$ or $\alpha = \frac{3\pi}{4} + n\pi$, for $n \in \mathbb{Z}$.
 (D) $\alpha = \frac{\pi}{4} + 2n\pi$ or $\alpha = \frac{3\pi}{4} + 2n\pi$, for $n \in \mathbb{Z}$.

8 A bag contains 5 identical blue marbles, 6 identical black marbles and 3 identical red marbles. Three marbles are drawn at random. Which expression below gives the correct probability that exactly two blue marbles are drawn?

- (A) $\frac{{}^3C_2}{{}^{14}C_3}$
 (B) $\frac{{}^5C_2 \times {}^9C_1}{{}^{14}C_3}$
 (C) $\frac{2 \times \frac{1}{9}}{5 \times \frac{1}{9}}$
 (D) $\frac{2 \times \frac{1}{14}}{5 \times \frac{1}{9}}$

$\frac{{}^5C_2 \times {}^9C_1}{{}^{14}C_3}$

B

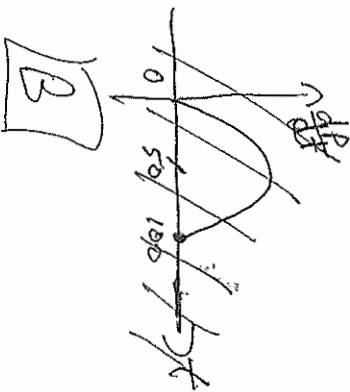
9 If $f(x) = \frac{3+e^{2x}}{5}$, which of the following is $f^{-1}(x)$?

- (A) $\ln(5x-3)$
 (B) $\frac{1}{2}\ln(5x-3)$
 (C) $\ln(5x)-\ln(3)$
 (D) $\frac{1}{2}(\ln(5x)-\ln(3))$
- $y = \frac{3+e^{2x}}{5}$
 $5y = 3+e^{2x}$
 $5y-3 = e^{2x}$
 $2x = \ln(5y-3)$
 $x = \frac{1}{2}\ln(5y-3)$
- [B]

10 The population, P , of animals in an environment in which there are scarce resources is increasing such that $\frac{dP}{dt} = P(100-P)$, where t is time. The initial population is 20 animals. Which of the following is true?

- (A) $P = 100 - 80e^{100t}$
 (B) The population is increasing most rapidly when $P = 50$.
 (C) The population is increasing most rapidly when $t = 50$.
 (D) The maximum population is $P = 50$.

End of Section 1



Q.11

(a) $x+y-5=0$, $x-2y-5=0$

$m_1 = -1$

$m_2 = \frac{1}{2}$

$\therefore$ Lines are perpendicular ✓

$= \left| \frac{-1 - \frac{1}{2}}{1 - (-\frac{1}{2})} \right|$

$= \left| \frac{-\frac{3}{2}}{\frac{1}{2}} \right|$

$= |-3|$

$\alpha = 71.34^\circ$ ✓

(b)

$\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \cos^2 x \, dx$

$= \frac{1}{2} \int_{\frac{\pi}{16}}^{\frac{\pi}{8}} (1 + \cos 2x) \, dx$ ✓

$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$ ✓

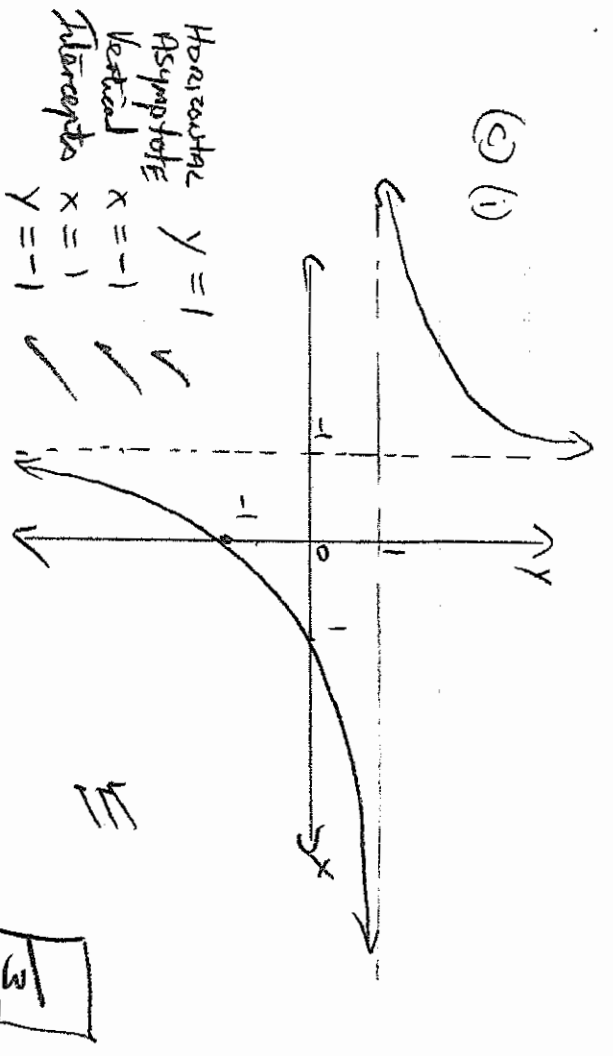
$= \frac{1}{2} \left[\left(\frac{\pi}{8} + \frac{1}{4} \sin \frac{\pi}{4} \right) - \left(\frac{\pi}{16} + \frac{1}{4} \sin \frac{\pi}{8} \right) \right]$

$= \frac{1}{2} \left[\frac{\pi}{8} + \frac{1}{4} - \frac{\pi}{16} - \frac{1}{4} \times \frac{1}{\sqrt{2}} \right]$

$= \frac{1}{2} \left[\frac{\pi}{16} + \frac{1}{4} - \frac{1}{4\sqrt{2}} \right]$ ✓

$= \frac{1}{8} \left[\frac{\pi}{4} + 1 - \frac{1}{\sqrt{2}} \right]$

[3]



(ii) $\frac{x-1}{x+1} > 1$

Solution $x < -1$

2

(d) $3x^3 + 6x^2 - 5x - 8 = 0$

Sum $a+b+c = -\frac{6}{3} = -2$

Sum of products $ab+ac+bc = -\frac{5}{3}$

Product $abc = -\frac{8}{3}$

$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc+ac+ab}{abc} = \frac{-\frac{5}{3}}{-\frac{8}{3}} = \frac{5}{8}$

(e) $\sin 2\theta$ where $\theta = \tan^{-1} \frac{1}{2}$

$\tan \theta = \frac{1}{2}$

$\sin 2\theta = 2 \sin \theta \cos \theta$

$\sin \theta = \frac{1}{\sqrt{5}}$, $\cos \theta = \frac{2}{\sqrt{5}}$

$\sin 2\theta = 2 \cdot \frac{1}{\sqrt{5}} \cdot \frac{2}{\sqrt{5}} = \frac{4}{5}$

2

Q12

(a) $(2x^2 + \frac{1}{x})^9$

(i) $T_{r+1} = {}^9C_r (2x^2)^{9-r} \cdot (\frac{1}{x})^r$

$= {}^9C_r 2^{9-r} x^{18-2r} (-1)^r x^{-r}$

Let $18-3r = 6$

$3r = 12 \Rightarrow r = 4$

$\therefore \text{Coeff} = {}^9C_4 (2^5)$

$= 6032$

2

(ii) Consider $\frac{T_{r+1}}{T_r} > 1$

where $(2x^2 + \frac{1}{x})^9$

$\frac{{}^9C_{r+1} (2x^2)^{9-r-1} (\frac{1}{x})^{r+1}}{{}^9C_r (2x^2)^{9-r} (\frac{1}{x})^r} > 1$

$\frac{{}^9C_{r+1}}{{}^9C_r} \cdot \frac{(2x^2)^{9-r-1}}{(2x^2)^{9-r}} \cdot \frac{(\frac{1}{x})^{r+1}}{(\frac{1}{x})^r} > 1$

$\frac{9-r}{r+1} \cdot \frac{1}{2} \cdot \frac{1}{x} > 1$

$\frac{9-r}{r+1} > 2x$

$\frac{9-r}{r+1} > 2 \cdot \frac{1}{2} = 1$

$9-r > r+1$

$8 > 2r$

$r < 4$

$r = 3$

Use $\left[\frac{n-r+1}{r} \cdot \frac{b}{a} > 1 \right]$

$\left[\frac{{}^9C_r}{{}^9C_{r-1}} \cdot \left(\frac{1}{2} \right) \cdot \left(\frac{1}{x} \right)^{-1} \right]$

$\frac{9-r+1}{r} \cdot \frac{1}{2} \cdot x > 1$

$\frac{10-r}{r} \cdot \frac{1}{2} \cdot x > 1$

$\frac{10-r}{2r} > 1$

$10-r > 2r$

$3r < 10$

$r < \frac{10}{3}$

$r = 3$

$\therefore$ greatest Coeff

$\therefore \frac{{}^9C_3}{{}^9C_2} \times \frac{1}{2}$

$$12(b) \int_0^{25} x \sqrt{9-x} \, dx$$

$$u = 9-x$$

$$\frac{du}{dx} = -1$$

$$= \int_9^4 (9-u) \sqrt{u} \cdot -1 \, du$$

$$= \int_4^9 [9u^{\frac{1}{2}} - u^{\frac{3}{2}}] \, du$$

$$= \left[\frac{9u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{\frac{5}{2}}}{\frac{5}{2}} \right]_4^9$$

$$= \left[6u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}} \right]_4^9$$

$$= 6(9)^{\frac{3}{2}} - \frac{2}{5}(9)^{\frac{5}{2}} - 6(4)^{\frac{3}{2}} + \frac{2}{5}(4)^{\frac{5}{2}}$$

$$= \frac{148}{5}$$

///

$$(c) \text{ Ratio } k:1 \quad A(-3,6), B(7,-8)$$

$$\therefore \left[\frac{7k-3}{k+1}, \frac{-4k+6}{k+1} \right] = [1, 2]$$

$$\therefore \frac{7k-3}{k+1} = 1 \quad \frac{-4k+6}{k+1} = 2$$

$$7k-3 = k+1$$

$$6k = 4$$

$$k = \frac{2}{3}$$

$$-4k+6 = 2k+2$$

$$6k = 4$$

$$k = \frac{2}{3}$$

///

$$17(d) \quad T_A = 25 + 150 e^{-\left(\frac{1}{20} \ln \frac{3}{5}\right)t}$$

$$T_B = 25 + 250 e^{-kt}$$

$$(i) \text{ For } B \quad t=20 \quad T=175$$

$$175 = 25 + 250 e^{-kt}$$

$$250 e^{-kt} = 150$$

$$e^{-kt} = \frac{150}{250}$$

$$\therefore e^{-20k} = \frac{3}{5}$$

$$-20k = \ln\left(\frac{3}{5}\right)$$

$$k = \frac{1}{20} \ln \frac{5}{3}$$

///

$$(ii) \text{ At } 25 + 150 e^{-\left(\frac{1}{20} \ln \frac{3}{5}\right)t} = 25 + 250 e^{-\left(\frac{1}{20} \ln \frac{3}{5}\right)t}$$

$$150 e^{-\left(\frac{1}{20} \ln \frac{3}{5}\right)t} = 250 e^{-\left(\frac{1}{20} \ln \frac{3}{5}\right)t}$$

$$\therefore e^{-\left(\frac{1}{20} \ln \frac{3}{5}\right)t} + \left(\frac{1}{20} \ln \frac{3}{5}\right)t = \frac{250}{150}$$

$$\therefore t \left[\frac{1}{20} \ln \frac{3}{5} - \frac{1}{20} \ln \frac{3}{5} \right] = \ln \frac{5}{3}$$

$$t \times \ln \left(\frac{5}{3} \right) = \frac{100}{20} \times \ln \frac{5}{3}$$

$$t = \frac{100 \times 20 \ln \frac{5}{3}}{\ln \left(\frac{10}{9} \right)} = \underline{\underline{97 \text{ min}}}$$

$$n(e) \quad \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!} \quad n \geq 1$$

(i) Test for $n=1$

$$LHS = \frac{1}{2!} = \frac{1}{2}$$

$$RHS = \frac{2! - 1}{2!} = \frac{2 - 1}{2} = \frac{1}{2}$$

$\therefore$ True for $n=1$

(ii) Assume true for $n=k$

$$\therefore \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)! - 1}{(k+1)!} *$$

(ii) Test for $n=k+1$

$$\therefore LHS = \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = \frac{(k+2)! - 1}{(k+2)!} *$$

Now LHS = $\frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$ *from equation **

MARKING

1- Step (i) + Conclusion

2- Step (ii) + (iii)

$$= \frac{((k+1)! - 1)(k+2) + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - k - 2 + k + 1}{(k+2)!} \quad \text{as } (k+1)!(k+2) = (k+2)!$$

$$= \frac{(k+2)! - 1}{(k+2)!} = RHS$$

$\therefore$ if true for $n=k$ hence true for $n=k+1$

But as it is true for $n=1$ it must be true for $n=2, 3, \dots$ and hence true for all n , by Mathematical Induction

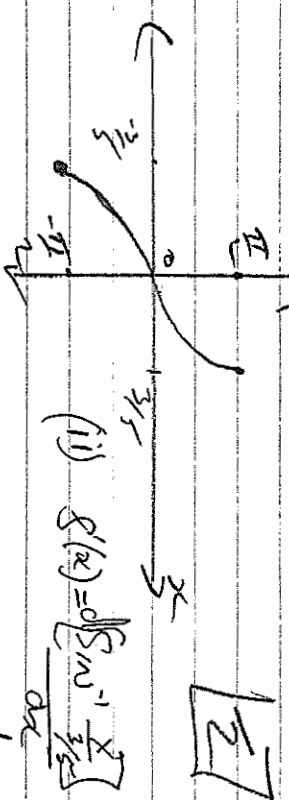
Q.13

(a) (i) $g(x) = \sin^{-1}(\frac{5x}{3})$

Domain: $-1 \leq \frac{5x}{3} \leq 1$

$-\frac{3}{5} \leq x \leq \frac{3}{5}$

Range: $-\frac{\pi}{2} \leq g(x) \leq \frac{\pi}{2}$



(b) $\int_0^{\sqrt{3}} \frac{2}{9+4x^2} dx$

$$= \frac{1}{9} \int_0^{\sqrt{3}} \frac{2}{1 + (\frac{2x}{3})^2} d(\frac{2x}{3})$$

$$= \frac{1}{9} \left[\tan^{-1} \frac{2x}{3} \right]_0^{\sqrt{3}}$$

$$= \frac{1}{9} \left[\tan^{-1} \frac{2\sqrt{3}}{3} - \tan^{-1} 0 \right]$$

$$= \frac{1}{9} \left[\tan^{-1} \frac{2\sqrt{3}}{3} \right]$$

✓

✓

✓

✓

✓

✓

$$13(ii) \rho(\cos 45^\circ \text{ max}) = 20 C_8 \left(\frac{1}{3}\right)^6 \left(\frac{4}{3}\right)^{14}$$

$$= 0.00000281474$$

$$(ii) \left(\frac{1}{3}\right)^6 \times \left(\frac{4}{3}\right)^{14}$$

check

(14)



(i) From Ref Sheet Normal:

$$x + py = ap^3 + 2ap$$

$$\text{let } x=0 \therefore y = ap^3 + 2a$$

$$N(0, ap^3 + 2a)$$

$$(ii) M \left[\frac{0+2ap}{2}, \frac{ap^3+2ap}{2} \right]$$

$$= (ap, ap^2 + a)$$

(iii) locus

$$x = ap$$

$$\therefore p = \frac{x}{a}$$

$$y = ap^2 + a$$

$$y = a \left(\frac{x}{a} \right)^2 + a$$

$$y - a = \frac{x^2}{a}$$

$$Q14(a) \quad 3 \sin x + \sqrt{3} \cos x = R \sin x \cos x + R \cos x \sin x$$

$$\therefore R \cos x = 3 \quad \therefore (i)$$

$$R \sin x = \sqrt{3} \quad \therefore (ii)$$

$$\therefore (i)^2 + (ii)^2 \Rightarrow R^2 (\sin^2 x + \cos^2 x) = 12$$

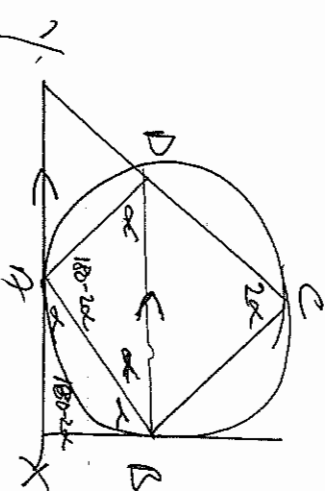
$$R = \sqrt{12}$$

(ii) (i)

$$\tan \alpha = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$

(b)



$$(i) \angle BDA = \alpha \quad [\text{Alt } \angle \text{ Theorem}]$$

$$\angle DBA = \alpha \quad [\text{Alt } \angle, \text{ DE } \parallel \text{YC}]$$

$$\therefore \angle DAB = 180 - 2\alpha \quad [\text{sum } \Delta]$$

$$\therefore \angle BCD = 2\alpha \quad [\text{ABCD is cyclic Quad}]$$

$$(ii) \text{ IN } \Delta AXB, \quad AX = BX \quad [\text{TRISECT LINE}]$$

$$\therefore \angle ABX = \alpha$$

$$\therefore \angle AXB = 180 - 2\alpha$$

$$\therefore \angle BXC \text{ is a cyclic Quad}$$

$$\therefore \angle BCA = 2\alpha$$

14 (C)

(i) A B C D E F

No. of letter with A = $1 \times 5 \times 4 \times 3$

$$= 60$$

(ii) $\frac{6 \times 5 \times 4}{2}$

$$= 6$$

or 6 ways to select letter

or $6 \times 5 \times 4 \times 3 = 120 \times 6 = 720$

(d)(i) $y = \tan \theta - \frac{g x^2}{2v^2} \sec^2 \theta$

let $y=0$ then $\frac{g x^2}{2v^2} \sec^2 \theta = 0$

$$\frac{g \sec^2 \theta}{2v^2} \cdot x^2 = \tan \theta$$

$$x^2 = \frac{2v^2 \tan \theta}{g \sec^2 \theta}$$

$$= 2v^2 \cdot \frac{\sin \theta}{\cos \theta} \cdot \cos^2 \theta$$

$$x^2 = \frac{v^2 \sin 2\theta}{g}$$

$$R = \frac{v^2 \sin 2\theta}{g}$$

$$R = x^2$$

(ii) $R_1 = x - 100$ when $\theta = 15^\circ$ $R_2 = 558.8 + x$ $\theta = 30^\circ$

$$\frac{v^2 \sin 30^\circ}{g} = x - 100 \quad \frac{v^2 \sin 60^\circ}{g} = 558.8 + x$$

$$\frac{x - 100}{\sin 30^\circ} = \frac{558.8 + x}{\sin 60^\circ}$$

$$\frac{\sqrt{3}}{2} (x - 100) = \frac{1}{2} (558.8 + x)$$

$$\sqrt{3}x - x = 658.8$$

$$x = \frac{658.8}{\sqrt{3} - 1}$$

$$= 899.96$$

from *

$$\frac{v^2}{g} = \frac{558 + x}{\sqrt{3}}$$

$$= \frac{558 + 899.96}{\sqrt{3}}$$

$$v^2 = 1683.48$$

Now

$$\therefore \sin 2\theta = \frac{1683.48}{1683.48} \quad 2\theta = \sin^{-1}(1) = 180^\circ$$